UTG 白皮书

0. 封面（Cover Page）

UTG · Urban Treasure Gold
基于墨西哥电子废料实业的 RWA 资产收益型通缩代币
UTG · Urban Treasure Gold
A Deflationary RWA Token Backed by Real Electronic-Waste Industrial Cashflow in Mexico
版本：v1.1（2025）
Version: v1.1 (2025)

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1. 项目概述（Project Overview）

1.1 行业背景（Industry Background）

- 墨西哥及拉美地区电子废料、芯片、电路板与稀有金属回收行业快速增长。
  Mexico and Latin America are experiencing rapid growth in e-waste, chips, PCB and precious metal recycling industries.
- 行业内现金流需求大、回款周期 30–120 天不等。
  The industry requires continuous cashflow with repayment cycles of 30–120 days.
- 传统跨境贸易在资金结算、外汇、合规成本方面效率低、周期长、成本高。
  Traditional cross-border trade suffers from slow settlement, forex friction, high compliance cost, and operational delays.
- 投资端缺乏具备真实资产支撑、现金流分红与透明账本的长期标的。
  Investors lack long-term products backed by real assets, real cashflow, and transparent accounting.
- 区块链可提供：全球资金接入、智能合约自动处理、跨境实时结算、透明规则。
  Blockchain enables global capital access, smart-contract automation, real-time settlement, and transparent rules.

1.2 UTG 的定位（UTG Positioning）

UTG 以墨西哥电子材料回收产业链为基础，将真实现金流与区块链通证经济结合，构建一个 **可持续 10 年以上的资产收益型 Token 体系**：
UTG is built on Mexico’s real electronic-waste industrial chain, combining real-world cashflow with blockchain token economics to create a sustainable, 10-year+ yield-backed token system.

- UTG 是整个系统的价值载体与治理代币；
  UTG is the value and governance token of the ecosystem.
- DApp 作为“资产上链平台”，承载活期池、定期池、任务池；
  The DApp acts as the on-chain asset platform integrating flexible, fixed, and project-based pools.
- 真实项目净利润按季度用于回购销毁 UTG，形成长期通缩机制。
  Real industrial profits are used quarterly to buy back and burn UTG, forming long-term deflation.

1.3 核心结构（Core Architecture）

本系统包含三大模块：
The system consists of three primary modules:

1. **活期诸蓄池（Flexible Pool）**：灵活存取、实时计息、目标 3–8% 年化区间。
   Flexible Pool: Free deposits/withdrawals, real-time interest, target 3–8% APY range.
2. **定期诸蓄池（Fixed Pool）**：30/90/180/365 天锁仓，对应固定收益区间。
   Fixed Pool: Lock periods of 30/90/180/365 days with fixed APR tiers.
3. **项目任务池（Project Task Pool）**：真实墨西哥实业项目的专项投资池。
   Project Task Pool: Dedicated pools linked to real industrial projects in Mexico with profit-sharing settlement.

1.4 合规与安全框架（Compliance & Security Framework）

UTG 在设计与运营中坚持遵循国际通用的数字资产合规要求，确保项目能够在真实世界业务与链上金融之间保持稳健、透明、可持续的发展路径。
UTG adheres to international standards for digital-asset compliance, ensuring stability, transparency, and sustainability across real-world operations and on-chain financial activities.

1.4.1 合规方向说明（Regulatory Compliance Statement）

UTG 将根据主要司法辖区（包括但不限于美国、欧盟、墨西哥、亚洲市场等）的数字资产法规开展业务。
UTG operates in accordance with digital-asset regulations across major jurisdictions, including the U.S., EU, Mexico, and Asian markets.

项目不向法律禁止或限制参与数字资产交易的地区主动进行营销或推广。
The project does not actively promote or offer tokens in regions where digital-asset participation is restricted or prohibited.

UTG Token 不构成证券、不代表股权、不赋予未来收益承诺，仅作为生态使用与治理用途。
UTG Token is not a security, does not represent equity, and does not guarantee future returns; it is used solely for ecosystem utility and governance.

实业项目的现金流参与与利润分配项，通过链上智能合约规则运行，不构成任何形式的收益保证。
All yield-related mechanisms are governed strictly by smart-contract rules and real industrial performance, without any form of guaranteed return.

1.4.2 智能合约审计计划（Smart Contract Audit Plan）

为确保系统安全性、透明度与稳定性，UTG 将在主网上线前后委托第三方安全机构进行全面智能合约审计。
UTG will engage reputable third-party security firms to perform comprehensive smart-contract audits before and after mainnet deployment.

审计范围包含：UTG Token、Flexible Pool、Fixed Pool、Project Pool、Treasury、BuybackBurn 模块。
The audits will cover all major modules: UTG Token, Flexible Pool, Fixed Pool, Project Pool, Treasury, and BuybackBurn contracts.

审计完成后将公开发布审计报告，供社区和投资人查阅。
Audit reports will be published publicly to ensure transparency for the community and investors.

后续重大版本更新亦将进行二次审计或增量审计。
Major updates will undergo follow-up or incremental audits to maintain long-term system security.

1.4.3 资金托管与透明度（Fund Custody & Transparency Model）

UTG 重视资产透明度与用户资金安全，将采用链上公开、可验证的方式管理项目池资金。
UTG prioritizes transparency and user fund safety by implementing on-chain, publicly verifiable fund management.

核心措施包括：
Core measures include:

关键项目池公开链上地址
所有活期池、定期池、项目任务池均提供公开可查询的链上地址，实现全程透明。
All pool addresses (flexible, fixed, project pools) are fully transparent and verifiable on-chain.

多签钱包 / 半托管结构
项目金库（Treasury）、回购资金池等关键资产将采用多签（Multi-Sig）或托管方案，提升资金安全性，防止单点权限风险。
Treasury and buyback funds may use multi-signature or custodial structures to eliminate single-point risks.

实时区块链可验证账本
所有收益分配、回购行为、销毁动作均在链上记录，可由任何用户验证。
All yield distributions, buyback actions, and token burns are recorded on-chain for full public verification.

实业项目回款透明化流程
项目收益注入到任务池前，将通过管理员端与链上合约双重确认，避免人为误差与操作风险。
Industrial profit settlements undergo dual verification (admin + contract) before being injected into project pools to avoid operational risk.

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2. 产品整体结构（Product Structure）

UTG 的产品由 **三类资金池** 组成，用于将不同风险偏好、不同周期的资金需求进行系统化管理：
UTG consists of three types of pools designed to allocate capital according to user risk preference and holding period.

2.1 活期诸蕃池（Flexible Pool — Liquidity Yield Layer）

面向希望保持高流动性与低风险的用户，支持随存随取，收益由实业利润 + 平台手续费动态支持。
Designed for users requiring high liquidity and low risk. Flexible deposits and withdrawals. Yield comes from real-world business profits + platform fees.

2.2 定期诸蓄池（Fixed Pool — Fixed Yield Layer）

面向愿意锁定一定周期以换取稳定目标收益的用户，30/90/180/365 天对应不同的预期年化与权重。
Designed for users willing to lock funds for stable target yields. Lock periods of 30/90/180/365 days correspond to different fixed APR levels and weight multipliers.

2.3 项目任务池（Project Task Pool — Real-World Business Yield Layer）

对接真实墨西哥电子废料与金属回收项目，单个项目专款专用，按利润比例进行结算分成。
Connected to real Mexican industrial projects (e-waste, chips, metals). Each pool is project-specific, with returns distributed based on actual profit.

三层收益结构（Three-Layer Yield Architecture）

流动性收益层（活期） → 固定收益层（定期） → 实业驱动收益层（任务池）
Liquidity Yield → Fixed Yield → Real-World Business Yield

完整覆盖不同类型用户的资金需求，并将墨西哥实业的真实现金流带入 UTG 生态。
Together, they cover all user capital preferences while channeling real Mexican industrial cash flow into the UTG ecosystem.

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3. 代币经济模型（Tokenomics）

3.1 基本信息（Basic Information）

- Token 名称：**UTG（Urban Treasure Gold）**
  Token Name: UTG (Urban Treasure Gold)
- 公链：**BSC**
  Blockchain: BSC
- 总供应量：**100,000,000 UTG（永不增发）**
  Total Supply: 100,000,000 UTG (No further issuance; fixed supply)

3.2 代币分配结构（Token Distribution）

分类	占比	数量	备注
私募（Seed + Private）	15%	15,000,000	价格 0.06–0.075 USDT，6 个月锁仓 + 12 个月线性解锁
Private Sale (Seed + Private)	**15%**	**15,000,000**	**Price 0.06–0.075 USDT; 6-month lock + 12-month linear release**
初始 LP	12–15%	12–15M	配对 120–150 万 USDT，LP 锁 24 个月
Initial LP	**12–15%**	**12–15M**	**Paired with 1.2–1.5M USDT; LP locked 24 months**
团队 / 顾问	10%	10,000,000	12 个月锁 + 36 个月线性解锁

分类	占比	数量	备注
Team / Advisors	**10%**	**10,000,000**	**12-month lock + 36-month linear release**
生态 / 社区激励	20%	20,000,000	空投、激励、合作伙伴奖励
Ecosystem / Community Incentives	**20%**	**20,000,000**	**Airdrops, rewards, partner incentives**
回购销毁池	15–20%	15–20M	用于实业利润回购并销毁
Buyback & Burn Pool	**15–20%**	**15–20M**	**Funded by real-world industrial profits**
市场化流动性支持	3–5%	3–5M	做市、跨链桥
Market Liquidity Support	**3–5%**	**3–5M**	**Market-making, cross-chain liquidity**
项目金库与储备池	15–25%	15–25M	安全垫与未来扩展
Treasury & Reserves	**15–25%**	**15–25M**	**Buffer fund and future expansion**

3.3 长周期释放与通缩模式（Long-Term Emission & Deflation Model）

- **前 3 年：高强度社区激励与锁仓激励。**
  Years 1–3: High-intensity community incentives and lock-based rewards.

- **第 4–6 年：奖励减半，生态进入稳定增长阶段。**
  Years 4–6: Reward reduction (halving phase), ecosystem grows with real-world usage.

- **第 7–10 年：削减代币增发，主要由实业利润驱动回购销毁。**
  Years 7–10: Minimal token emissions; deflation driven mainly by industrial profit buybacks.

目标：10 年累计销毁 15–20% 总供应量。
Target: Reduce total supply by 15–20% within 10 years through systematic buyback & burn.

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4. 模块功能详解（Modules）

4.1 活期诸蕃池（Flexible Pool）

功能（Functions）

- 支持 UTG / USDT 存入；
  Supports deposits in UTG / USDT.
- 存入当日开始计息；
  Interest starts accruing on the same day of deposit.

- 随时赎回，本金 + 累计收益实时计算；
  Funds can be withdrawn anytime with real-time calculation of principal + accrued yield.
- 目标年化收益区间：**3%–8%**。
  Target APY range: 3%–8%.

用户流程（User Flow）

1. 选择币种（UTG 或 USDT）；
   Select token (UTG or USDT).
2. 输入存入金额 → 点击「存入」；
   Enter deposit amount → Click “Deposit”.
3. DApp 记录活期池份额；
   DApp records user’s share in the flexible pool.
4. 当天计息；
   Interest begins the same day.
5. 用户可随时点「提取」赎回。
   Users may click “Withdraw” at any time.

前端展示（Frontend Display）

- 当前 APR（如：5.2%）；
  Current APR (e.g., 5.2%).
- 用户累计收益；
  Total accumulated yield.
- 每日预估收益；
  Estimated daily earnings.
- 当前余额、收益进度、交易记录。
  Current balance, yield progress, and transaction history.

4.2 定期诸蓄池（Fixed Pool）

锁仓档位（固定 APR）

Lock Durations (Fixed APR)

锁仓天数	年化收益	**Lock Days**	**APR**
30 天	8%	**30 days**	**8%**
90 天	10%	**90 days**	**10%**
180 天	12%	**180 days**	**12%**
365 天	14%	**365 days**	**14%**

用户流程（User Flow）

1. 选择锁仓金额与天数；
   Select the staking amount and lock duration.

2. 锁仓后显示：开始时间 / 预计到期时间；
   The interface shows the start date and estimated maturity date.
3. 到期后按钮变为「可赎回」，用户点击领取；
   When maturity is reached, the button becomes “Redeem”, allowing the user to claim funds.
4. 本金 + 全额利息返还；
   User receives 100% principal + full interest.
5. 如提前赎回：
   If redeemed early:
6. 领到 **30% 已产生利息**；
   User receives 30% of the interest generated so far.
7. **70% 回流奖励池**（项目方指定钱包）。
   The remaining 70% is returned to the Reward Pool (project-designated wallet).

前端展示（Frontend Display）

- 锁仓金额
  Locked amount
- 锁仓天数
  Lock duration
- 锁仓开始/结束时间
  Start / Maturity timestamps
- 到期收益
  Maturity yield
- 进度条与倒计时
  Progress bar and countdown timer

4.3 项目任务池（Project Task Pool）

任务池由真实墨西哥实业项目驱动（Project Pools Are Driven by Real Mexican Industrial Projects）

示例任务：
Example Task:

- **项目名称**：电子废料回收项目（一期）
  Project Name: E-waste Recycling Project (Phase 1)
- **目标资金**：200,000 USDT
  Target Funding: 200,000 USDT
- **预计周期**：120 天
  Estimated Duration: 120 days
- **预计利润率**：20%（示例，不构成保证）
  Estimated Profit Margin: 20% (example only, not a guarantee)
- **利润分配**：参与者 15%，平台 5%
  Profit Sharing: 15% to participants, 5% to platform

前端展示（Frontend Display）

- 项目名称
  Project name
- 目标金额 / 已筹金额
  Target amount / Amount raised
- 进度条
  Progress bar
- 预计周期
  Estimated duration
- 预计利润率
  Estimated profit margin
- 分配比例（如参与者 15%）
  Profit distribution (e.g., 15% to participants)
- 项目状态（募集中 / 进行中 / 已结束 / 已结算）
  Project status (Funding / Ongoing / Completed / Settled)

参与流程（Participation Flow）

1. 选择项目；
   Select a project.
2. 选择币种 UTG / USDT；
   Choose token UTG / USDT.
3. 资金锁定至项目结束；
   Funds are locked until project completion.
4. 状态变为「可结算」后，管理员注入实际回款；
   Once marked “Settleable”, admin injects actual project returns.
5. 用户点击“领取” → 获得本金 + 分成收益。
   User clicks “Claim” → Receives principal + profit share.

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5. 基础公共功能（Common Functions）

5.1 用户端（User Interface）

- 用户总资产（活期 + 定期 + 项目池）；
  Total user assets (Flexible + Fixed + Project Pools).
- 每日收益计算；
  Daily yield calculation.
- 历史收益明细；
  Historical earnings records.
- 任务池参与记录；
  Participation history for project task pools.
- 多币种余额显示。
  Multi-currency balance display.

5.2 管理员端（Admin Interface）

- 调整活期 APR（3–8% 区间内）；
  Adjust flexible pool APR (within 3–8% range).
- 创建/编辑任务池（目标金额、预计周期、分配比例等）；
  Create/Edit project task pools (target amount, duration, profit allocation).
- 执行任务池结算，并注入实际利润；
  Execute project settlement and inject actual profit returns.
- 管控奖励池与销毁池相关操作。
  Manage reward pool and burn pool operations.

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6. 回购与销毁机制（Buyback & Burn）

回购来源（Sources of Buyback Funds）

- 墨西哥实业净利润的一部分；
  A portion of net profits from Mexican industrial operations.
- 平台手续费；
  Platform fees generated by the ecosystem.
- 奖励池节余资金。
  Unused or surplus funds from the reward pool.

回购方式（Buyback Methods）

- 按季度进行；
  Executed on a quarterly basis.
- 在 DEX/CEX 分批回购；
  Buybacks conducted in batches on DEX/CEX.
- 统一打入黑洞地址销毁。
  All repurchased UTG is sent to a burn (black-hole) address.

长期效果（Long-Term Effects）

- 流通筹码逐年减少；
  Circulating supply decreases year by year.
- 实业规模越大，回购力度越强；
  The larger the industrial operations, the stronger the buyback power becomes.
- 强化 UTG 长期价值与通缩预期。
  Strengthens UTG’s long-term value and enhances market deflation expectations.

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7. 路线图（Roadmap）

● 第 1–2 年（启动期）

Years 1–2 (Launch Phase)

- 推出 UTG Token；
  Launch UTG Token.
- DApp v1（活期/定期上线）；
  Release DApp v1 (Flexible & Fixed Pools online).
- 落地墨西哥 1–3 个项目示范；
  Launch 1–3 pilot industrial projects in Mexico.
- 完成第一轮回购销毁。
  Complete first round of buyback & burn.

● 第 3–5 年（扩张期）

Years 3–5 (Expansion Phase)

- DApp v2：多国家、多项目并行；
  Deploy DApp v2: multi-country, multi-project support.
- 增加不同类型实业资产（电子、金属、能源等）；
  Add additional industrial asset categories (electronics, metals, energy, etc.).
- UTG 上线至少一家 CEX；
  List UTG on at least one centralized exchange (CEX).
- 扩大金库规模。
  Expand treasury reserves.

● 第 6–10 年（平台化时期）

Years 6–10 (Platform Maturity Phase)

- 打造成熟的“拉美 RWA 实业收益平台”；
  Develop UTG into a mature Latin-American RWA industrial yield platform.
- 回购销毁累计目标 15–20%；
  Achieve cumulative buyback & burn target of 15–20%.
- UTG 成为全球配置墨西哥实业的价值入口。
  Position UTG as a global gateway to Mexican industrial investment.

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8. 风险提示（Risk Disclosure）

- 市场风险：代币价格波动不可避免；
  Market Risk: Token price volatility is unavoidable.
- 实业风险：原材料、回收价格波动可能影响收益；
  Industrial Risk: Raw material and recycling price fluctuations may affect returns.
- 汇率风险：USDT 与本地货币之间存在浮动；
  Forex Risk: Exchange rate fluctuations between USDT and local currencies.
- 监管风险：各国对代币与数字资产监管不一致；
  Regulatory Risk: Regulations on tokens and digital assets vary across countries.

- 技术风险：智能合约虽审计，但仍有潜在风险。
  Technical Risk: Smart contracts, although audited, may still contain vulnerabilities.

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9. 团队信息（Team Information）

- 团队在电子废料、芯片、电路板回收行业深耕多年；
  The team has many years of experience in e-waste, chip, and PCB recycling industries.
- 在墨西哥拥有长期运营团队与供应链体系；
  We maintain long-term operational teams and supply-chain networks in Mexico.
- 在区块链、智能合约开发方面具备专业经验。
  We possess strong expertise in blockchain and smart-contract development.

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10. 法律声明（Legal Disclaimer）

- 本白皮书不构成投资建议；
  This whitepaper does not constitute investment advice.
- UTG 不承诺固定收益或保证回报；
  UTG does not guarantee fixed returns or profits.
- 所有示例收益率仅为目标或估算；
  All yield examples are targets or estimates only.
- 投资需用户自行判断与承担风险。
  Users must evaluate risks and make investment decisions independently.

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11. 系统架构与流程图（Architecture & Flow Diagrams）

以下为 UTG 生态的核心 ASCII 图示，便于阅读、展示与放入 Word / PDF：

11.1 UTG 系统总体架构图（UTG System Architecture）

```
                  +-------------------------+
                  |      用户 / Users       |
                  |  UTG / USDT 资金进入     |
                  +-----------+-------------+
                              |
                              v
                +-------------+--------------+
                |     UTG DApp 平台          |
                |  UTG DApp Platform         |
                +-------------+--------------+
                              |
        +----------+-----------+-----------+----------+
        |          |                       |          |
        v          v                       v          v
+---------------+ +----------------+ +----------------+ +-------------+
```

```
| 活期诸蓄池    | | 定期诸蓄池     | | 项目任务池     | | 奖励池/金库   |
| Flexible Pool | | Fixed Pool    | | Project Pools | | Treasury    |
+-------+-------+ +-------+--------+ +-------+--------+ +-------------+
        |                |                 |
        |                |                 v
        |                |        +----------------------+
        |                |        | 墨西哥实业项目        |
        |                |        | Mexico Industrial Ops |
        |                |        +----------+-----------+
        |                |                   |
        |                |                   v
        |                |        实际利润 / Real Profit
        |                |                   |
        v                v                   v
+-------------------------------------------------------------------+
|                    回购 UTG → 销毁（通缩）                         |
|             Buyback UTG → Burn (Deflation Mechanism)               |
+-------------------------------------------------------------------+
```

11.2 三大资金池工作流程（Three-Pool Workflow）

```
用户资金 → 选择池子 → 资金进入对应池子 → 开始计息/锁仓/项目执行
    |            |                |                    |
    v            v                v                    v
Flexible Pool  Fixed Pool     Project Pool       实业项目执行
    |            |                |                    |
    v            v                v                    v
随时取出       到期取出         项目结束结算         实际利润返还
    \            |                |                    /
     \           |                +---------+-----------+
      \          |                          |
       \         |                          v
        +---------+-----------------> 奖励池 / 金库 (Treasury)
```

11.3 UTG 价值流转图（UTG Value Flow Cycle）

```
+-------------------------+
|  用户购买 / 质押 UTG     |
| User Buys / Stakes UTG |
+-----------+-------------+
            |
            v
+-------------------------+
|   UTG 流入生态资金池     |
| UTG enters Pools       |
+-----------+-------------+
```

```
            |
            v
+-------------------------+
|  墨西哥实业使用资金       |
| Mexico Industrial Ops  |
+-----------+-------------+
            |
   实业产生真实利润
   Real-World Profit
            |
            v
+-------------------------+
|   利润的一部分用于回购     |
|  Part of Profit → Buyback |
+-----------+-------------+
            |
            v
+-------------------------+
|   回购的 UTG 全部销毁      |
| Buyback UTG → Burn      |
+-----------+-------------+
            |
            v
+-------------------------+
|   市场流通量减少（通缩）     |
| Supply Decreases (Deflation) |
+-----------+-------------+
            |
            v
+-------------------------+
|   UTG 长期价值提升        |
| UTG Long-Term Value Up |
+-------------------------+
```

11.4 UTG 生态全景图（UTG Ecosystem Panorama）

```
                          +----------------------------+
                          |        全球用户 / 投资人       |
                          | Global Users & Investors    |
                          +----------------------------+
                                        |
                        资金 / Capital (USDT / UTG)
                                        |
                                        v
                          +----------------------------+
                          |        UTG DApp 前端        |
                          |     UTG DApp Frontend      |
                          +----------------------------+
                                        |
```

```
                                    v
              +------------------------+------------------------+
              |                        |                        |
              v                        v                        v
      +----------------+        +----------------+        +----------------+
      |  活期诸蓄池     |        |  定期诸蓄池     |        |  项目任务池     |
      | Flexible Pool |        |  Fixed Pool   |        | Project Pools  |
      +--------+-------+        +--------+------+        +--------+-------+
               |                        |                        |
               |                        |                        |
               |                 资金分配策略 / Yield Policy      |
               |                        |                        |
               v                        v                        v
      +-------------------------------------------------------------+
      |                    UTG 协议层 / 合约层                        |
      |            UTG Protocol / Smart Contracts                    |
      |  - 资金池合约  Pool Contracts                                 |
      |  - 任务池合约  Project Contracts                              |
      |  - 奖励池合约  Reward / Treasury Contracts                    |
      |  - 回购销毁合约 Buyback & Burn Contracts                      |
      +------------------------+------------------------+-------------+
                               |                        |
                               |                        |
                               v                        v
             +---------------------------+   +------------------------+
             | 墨西哥电子废料 & 金属实业运营 |   |   治理与风控 / Governance |
             | MX E-waste & Metal Ops    |   |   & Risk Management     |
             +------------+-------------+   +-------------+-----------+
                          |                               |
      供应链 / Supply      |                               |
      链路:                |                               |
电子废料 → 回收 → 拆解 → 分级 → 销售       |                               |
                          v                               |
               实际利润 / Real Profit                      |
                          |                               |
                          v                               |
             +---------------------------+                |
             |  回购 UTG → 销毁（通缩）     |                |
             |  Buyback UTG → Burn       |                |
             +------------+-------------+                |
                          |                               |
                          v                               |
             +---------------------------+                |
             | 二级市场 / 交易所          |<---------------+
             | DEX / CEX Markets        |
             +------------+-------------+
                          |
                          v
             +---------------------------+
             | UTG 二级价格 & 市值反馈     |
             | UTG Price & Market Cap    |
```

```
                +------------+-------------+
                             |
                             v
                +---------------------------+
                |   反馈至用户 / 投资人收益     |
                | Feedback to Users/Investors|
                +---------------------------+
```

11.5 DApp 功能架构图（DApp Functional & Technical Architecture）

```
+-------------------------------------------------------------+
|                    前端层 / Frontend Layer                   |
|                  Web / Mobile UTG DApp UI                   |
|  - 钱包连接：MetaMask / WalletConnect                        |
|  - 资产总览：UTG / USDT 余额、收益                            |
|  - 功能入口：活期池 / 定期池 / 项目任务池                      |
|  - 多语言界面：中文 / English                                 |
+------------------------------+------------------------------+
                               |
                               | HTTPS / JSON-RPC
                               v
+-------------------------------------------------------------+
|                API 网关层 / API Gateway Layer                |
|  - 路由转发：将请求分发到对应服务模块                          |
|  - 参数与签名校验                                             |
|  - 限流与安全（基础防刷、风控规则）                            |
|  - 日志与监控                                                 |
+------------------------------+------------------------------+
                               |
                               | 内部服务调用 / Service Calls
                               v
+-------------------------------------------------------------+
|           业务逻辑层 / Application Service Layer             |
|  - PoolService：活期 / 定期资金池业务逻辑                      |
|  - ProjectService：项目发布、参与、项目结束结算                 |
|  - UserService：用户持仓、收益统计、记录查询                    |
|  - AdminService：管理员参数配置、池子开关、结算操作             |
|  - OracleService（可选）：价格预言机、链上/链下数据             |
+------------------------------+------------------------------+
                               |
                               | Web3 SDK / RPC 调用          |
                               v
+-------------------------------------------------------------+
|           区块链接口层 / Web3 & Node Integration             |
|  - ethers.js / web3.js                                        |
|  - 连接 BSC RPC / 全节点                                      |
|  - 构造交易 → 用户本地签名 → 广播交易                          |
|  - 查询链上余额 / 事件 / 日志                                 |
+------------------------------+------------------------------+
```

```
                                |
                                | 调用智能合约 / Contract Calls
                                v
+-------------------------------------------------------------+
|               智能合约层 / Smart Contract Layer              |
|  - UGTToken: UTG 主代币合约 (BEP-20)                         |
|  - FlexiblePool: 活期池存取、份额计算、收益累计              |
|  - FixedPool: 定期锁仓、到期赎回、提前赎回罚息               |
|  - ProjectPool: 项目资金锁定、状态流转、项目分成结算         |
|  - RewardTreasury: 奖励池与金库管理                          |
|  - BuybackBurn: 回购记录、销毁地址管理                       |
+------------------------------+------------------------------+
                                |
                                | 区块链网络 / Blockchain      |
                                v
+-------------------------------------------------------------+
|                    BSC 区块链网络 / BSC Network              |
|  - 出块 / 打包交易                                           |
|  - 交易与状态不可篡改                                         |
|  - 全网节点验证合约执行结果                                   |
+-------------------------------------------------------------+
```